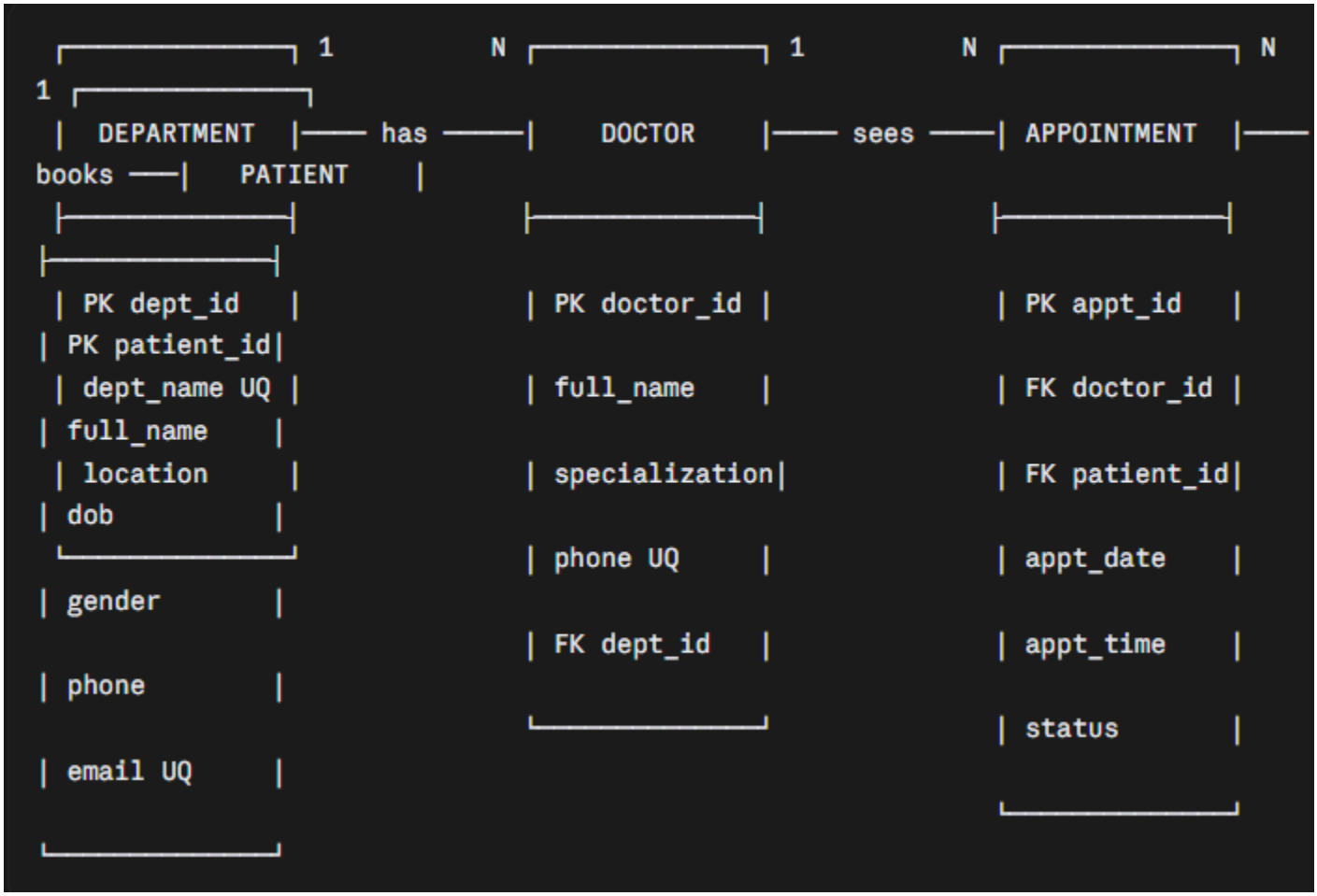
1. A hospital has doctors, patients, departments and appointments. A doctor belongs to one department. A patient can book many appointments. Draw the E-R diagram, create the tables with all keys and constraints, and insert 3 rows per table.



- A department has many doctors, and each doctor belongs to exactly one department (1:N).
- A patient can book many appointments (1:N).
- A doctor can have many appointments (1:N).
- Together, doctor and patient form a many-to-many relationship, and appointment is the junction table that resolves it.

```
CREATE DATABASE hospital_db;

USE hospital_db;
```

Department

```
CREATE TABLE department (
   dept_id   INT UNSIGNED NOT NULL AUTO_INCREMENT,
   dept_name VARCHAR(60) NOT NULL,
   location  VARCHAR(60),
   PRIMARY KEY (dept_id),
   UNIQUE KEY uq_dept_name (dept_name)
);
```

Doctor

```
CREATE TABLE doctor (
   doctor_id      INT UNSIGNED NOT NULL AUTO_INCREMENT,
   full_name      VARCHAR(100) NOT NULL,
   specialization VARCHAR(80),
   phone          VARCHAR(20) NOT NULL,
   dept_id        INT UNSIGNED NOT NULL,
   PRIMARY KEY (doctor_id),
   UNIQUE KEY uq_doctor_phone (phone),
   CONSTRAINT fk_doctor_dept FOREIGN KEY (dept_id) REFERENCES department (dept_id)
);
```

Patient

```
CREATE TABLE patient (
   patient_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
   full_name  VARCHAR(100) NOT NULL,
   dob        DATE NOT NULL,
```

```
    gender     ENUM('MALE','FEMALE','OTHER') NOT NULL,
    phone      VARCHAR(20) NOT NULL,
    email      VARCHAR(255),
    PRIMARY KEY (patient_id),
    UNIQUE KEY uq_patient_email (email),
    CONSTRAINT chk_patient_dob CHECK (dob <= '2100-01-01')
);
```

Appointment

```
CREATE TABLE appointment (
    appt_id    INT UNSIGNED NOT NULL AUTO_INCREMENT,
    patient_id INT UNSIGNED NOT NULL,
    doctor_id  INT UNSIGNED NOT NULL,
    appt_date  DATE NOT NULL,
    appt_time  TIME NOT NULL,
    status     ENUM('SCHEDULED','COMPLETED','CANCELLED') NOT NULL DEFAULT 'SCHEDULED',
    PRIMARY KEY (appt_id),
    CONSTRAINT fk_appt_patient FOREIGN KEY (patient_id) REFERENCES patient (patient_id),
    CONSTRAINT fk_appt_doctor  FOREIGN KEY (doctor_id)  REFERENCES doctor (doctor_id),
    CONSTRAINT uq_doctor_slot  UNIQUE (doctor_id, appt_date, appt_time)
);
```

```
INSERT INTO department (dept_name, location) VALUES
('Cardiology',  'Block A, Floor 2'),
('Pediatrics',  'Block B, Floor 1'),
('Orthopedics', 'Block A, Floor 3');

INSERT INTO doctor (full_name, specialization, phone, dept_id) VALUES
('Dr. Kamal Hossain', 'Cardiologist',          '01711000001', 1),
('Dr. Farhana Islam', 'Pediatrician',          '01711000002', 2),
('Dr. Mahbub Alam',   'Orthopedic Surgeon',    '01711000003', 3);

INSERT INTO patient (full_name, dob, gender, phone, email) VALUES
('Rahim Uddin', '1985-04-12', 'MALE',   '01811000001', 'rahim@example.com'),
('Salma Khatun','1992-09-30', 'FEMALE', '01811000002', 'salma@example.com'),
('Arif Chowdhury','2018-01-05','MALE',  '01811000003', NULL);

INSERT INTO appointment (patient_id, doctor_id, appt_date, appt_time, status) VALUES
(1, 1, '2026-11-10', '10:00:00', 'SCHEDULED'),
(2, 3, '2026-11-10', '11:30:00', 'SCHEDULED'),
(3, 2, '2026-11-05', '09:00:00', 'COMPLETED');
```

Section A

1. Define primary key and foreign key. Identify both in the appointment table.
2. Why is appointment called a junction (bridge) table? Which relationship does it resolve?
3. Write the difference between DELETE, TRUNCATE and DROP with one example each.
4. What is the purpose of the UNIQUE (doctor_id, appt_date, appt_time) constraint? What error occurs if it is violated?
5. What happens if you insert an appointment with doctor_id = 99 when no such doctor exists? Which constraint causes it?

Section B

6. Display each doctor's name with their department name (use INNER JOIN).
7. Display appt_id, patient name, doctor name, appt_date, appt_time and status for all appointments, ordered by date and then time.
8. Show the number of appointments for each doctor. Doctors with no appointments must also appear with 0 (use LEFT JOIN and COUNT).
9. List departments that have more than one doctor (GROUP BY and HAVING).
10. Find patients who have never booked an appointment. Write it in two ways: LEFT JOIN ... IS NULL and NOT IN (or NOT EXISTS).
11. Display the name and age (in years) of all patients older than 30 (use TIMESTAMPDIFF).
12. List all appointments in November 2026 for the Cardiology department (join three tables).
13. Using a subquery, list all doctors who work in the same department as "Dr. Kamal Hossain".
14. Change the status of all SCHEDULED appointments of patient 1 on a given date to CANCELLED. Run a SELECT check first, then the UPDATE.

Section C

15. Add a column fee DECIMAL(8,2) to doctor with a default of 500 and a check that it cannot be negative (ALTER TABLE).
16. Create a view upcoming_appointments that shows patient name, doctor name, date and time for appointments with status SCHEDULED. Then write a query to select from it.
17. Create an index on appointment (appt_date). Explain when an index helps and when it slows down inserts.
18. Draw the E-R diagram for this hospital, showing entities, attributes, primary keys, relationships and cardinality (1:N). Convert it into tables.

Section D

19. Explain INNER JOIN vs LEFT JOIN with an example from this database.
20. Why must WHERE be used before GROUP BY and HAVING after it? Give one query that uses both.